AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TT284
Web Technologies

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 16/April/2026

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: 60 **marks**
- This exam consists of **6 questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Basic Calculators** are allowed.

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PART 1: ALL QUESTIONS ARE REQUIRED [10 Marks]

Question 1: (10 marks)

1) The _____ led to the creation of the Transmission Control Protocol and the Internet protocols which now underpin all the everyday services of the internet, such as email, the Web, FTP, newsgroups, Web 2 technologies, and so on.
   A. CERN project.
   B. Internet Engineering Task Force project.
   C. Advanced Research Projects Agency Network.
   D. World Wide Web Consortium.

2) In the CSS, the _____ is contained within braces {} in the following form: { property1: value1; property2: value2; ... propertyN: valueN; }.
   A. selector.
   B. property.
   C. Metadata.
   D. declaration.

3) The _____ structure has different sections and topics, each topic is an island of information with all the pages in it are linked to each other. All pages contain clear navigation with links to the section pages, main page and site map.
   A. Hierarchical.
   B. Tutorial.
   C. Web.
   D. cluster.

4) The idea of creating _____ content is a simple one. So simple, that as a web designer, you really shouldn't be coding to exclude any of the disadvantaged visitors to your site.
   A. Dynamic.
   B. Interactive.
   C. usable.
   D. accessible.

5) When the browser does not support JavaScript or if JavaScript is turned off then content inside of _____ tags is displayed. This allows alternative 'non-JavaScript' content to be included in a page.
   A. noscript.
   B. JavaScriptOn.
   C. scriptenabled.
   D. scripton.

6) CSS _____.
   A. defines the structure of web pages using tags and elements.
   B. provides a simple mechanism for adding style in terms of fonts, colors, and positioning to the content of web documents.
   C. stores and retrieves data for dynamic websites.
   D. executes logic and handles user interactions in web applications.

7) When dealing with forms, the accept attribute is _____.
   A. a comma-separated list of content types the server will accept.
   B. a comma-separated list of character sets the form data may be in (in practice the server will accept the character set defined in the header of the document).
   C. a comma-separated list of content types the server should reject.
   D. the url that accepted to process the form data.

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8) Compiled languages are_____.
   A. translated to an executable file such as JavaScript, Basic, Perl, the complete program is translated before being executed. A major advantage of the compiled approach is that the program can be compiled once and run as many times as required without having to translate the high-level programming code into processing instructions each time the program is run.
   B. interpreted at execution time such as JavaScript, Basic, Perl, the key distinguishing property of interpreted languages is that they are interpreted from the programming language into processor instructions when the script is executed.
   C. translated to an executable file such as C, Delphi or Fortran, the complete program is translated before being executed. A major advantage of the compiled approach is that the program can be compiled once and run as many times as required without having to translate the high-level programming code into processing instructions each time the program is run.
   D. interpreted at execution time such as C, Delphi or Fortran, the key distinguishing property of interpreted languages is that they are interpreted from the programming language into processor instructions when the script is executed.

9) The navigator object _____.
   A. includes images object, which allows access to all the images that are included in the document.
   B. is a child of the window object, and allows us to manipulate a web page. For example, you can use the document object to insert content in a web page.
   C. is at the top level of the DOM hierarchy is a single window object, which represents the browser window. All the other objects in the DOM are contained directly or indirectly by the navigator object.
   D. allows us to retrieve information about the browser that has been used to access a page, you can then identify the name of the browser, and the operating system of the client.

10) require() function is used for including PHP files, _____.
   A. when an error is found in the file being included or if the file can't be found the function generates a fatal error, and the script will stop.
   B. when an error is found in the file being included or if the file can't be found, this function generates a warning, but the script will continue execution.
   C. when an error is found in the file being included or if the file can't be found the function will ask for fixing the error or select another directory otherwise the script will stop.
   D. when an error is found in the file being included or if the file can't be found, this function generates a warning, but the script will continue execution after fixing the error.

PART 2: ALL QUESTIONS ARE REQUIRED [20 Marks]

Question 2: (12 marks 3 marks each one)

a) List the three positioning modes available in CSS for placing elements within the box model.

b) There are a few key usability questions we should consider when creating a website, list three of them.

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c) Why should we be at all concerned about adhering to standards.
d) In HTML, the <img> element includes attributes that help address accessibility issues. Briefly explain these attributes.

Question 3: (8 marks 4 marks each one)

a) Explain how web applications split their computing workload.
b) Discuss the layers of Web Service Architecture.

PART 3: ALL QUESTIONS ARE REQUIRED [30 Marks]

Question 4: (5 marks)

Assume that a user is visiting a website with an average data transmission rate of 256 (Kbps); consider that the request has the following:

1) Four webpages (HTML tags) at a size of 1500 bytes.
2) Two CSS files at an average size of 3000 bytes for each file.
3) Three images at an average size of 1000 bytes for each file.
4) One banner image at a size of 2000 bytes.

Assume that it takes the server 3 milliseconds to respond to each file request. Calculate the total time required to download all these files.

Note:

1 Kbps = 1000 bits/second, 1 byte = 8 bits, 1 second = 1000 millisecond.

Question 5: (15 marks 5 marks each one)

a) Write the HTML code to create a sub ordered list that includes the following items in Figure 1:

1. Africa
    1. Nile
    2. Congo
2. Asia
    1. Yangtze
    2. Euphrates

Figure 1

b) What is the output of the following code?

```
<html>
<script>
 let x = 5;
 let output = "";
 let y = 1;
 while (y <= 5) {
  if ((x * y) % 4 == 0) {
    x -= 2;
  } else if (x % 2 == 0 || y % 2 !== 0) {
    x += 3;
  } else {
    x = x / 2;}
  output  += "X=" + x + ", Y=" + y + "<br>";
  y++; }
 document.write(output);
</script>  </html>
```

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c) Write a JavaScript program that performs the following tasks:
   1) Prompts the user to enter the names of 5 daily activities and the duration of each activity.
   2) Classifies each activity into one of the following categories:
      a. Quick Task (duration < 15 minutes)
      b. Focused Session (15 <= duration < 45 minutes)
      c. Extended Effort (45 <=duration < 90 minutes)
      d. Time-Intensive (duration >= 90 minutes)
   3) Print a summary showing how many activities fall into each category.

Question 6: (10 marks)

Develop a simple web page based on the following specification:

1. Page Title "Delivery Request".
2. Header 1 "Delivery Request Form".
3. Form action to locate the script that will process the form data. "http//norton.open.ac.uk/reflect.php".
4. Text box for Requester Name.
5. Text box for Requesrer email.
6. Checkbox for the "Express Delivery".
7. Checkbox for the "Home Delivery".
8. Upload payments receipt.
9. Submit button with Submit name.
10. Reset Button with Reset name.

The web page should look like the following design (Figure 2):



Figure 2

End of the questions.



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